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Primary Drivers of FDA Complete Response Letters: Insights from a SAS® -Based Analysis

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ABSTRACT

Complete Response Letters (CRLs) issued by the U.S. Food and Drug Administration (FDA) provide feedback when New Drug Applications or Biologic Licensing Applications contain deficiencies, deeming them unapprovable, but their unstructured format limits large-scale analysis. This project developed a SAS® -based workflow to convert CRLs into structured, analysis-ready datasets. CRLs were manually curated and coded into standardized deficiency domains and subtypes, and data were analyzed in SAS using PROC FREQ, co-occurrence summaries, and logistic modeling, with results visualized in an R Shiny dashboard. Among coded CRLs, Chemistry, Manufacturing, and Controls (CMC) deficiencies represented the largest domain (51.5%), followed by Clinical deficiencies (24.1%). Deficiency domain distributions differed significantly between NDA and BLA applications ($p < 0.001$), with BLAs showing a higher proportion of CMC-related deficiencies, and nearly 40% of CRLs involved multiple deficiency domains. This workflow shows how SAS can structure, analyze, and report complex regulatory data, providing insights into deficiency patterns to inform future drug development and support regulatory analytics – potentially reducing regulatory delays and costs.

INTRODUCTION

Complete Response Letters are issued by the U.S. Food and Drug Administration (FDA) when an application for a drug or biologic product cannot be approved with the current data. These letters provide regulatory feedback, including the reasons for non-approval and recommendations intended to strengthen the submission for future review. Deficiencies cited in CRLs generally fall within six major review domains: CMC, Clinical, Nonclinical, Regulatory, Statistics, and Patent infringement.

Given the number of CRLs issued over time, recurring deficiency trends likely exist across applications and product types. However, publicly available large-scale analyses are limited, in part because CRLs are released as unstructured narrative documents rather than analysis-ready datasets. As a result, systematic coding, aggregation, and statistical evaluation are difficult.

Because these patterns may help inform future development strategy and regulatory preparedness, publicly available CRLs were manually reviewed and coded into structured variables. Key application characteristics, deficiency domains, and subtype findings were captured using predefined classification criteria applied consistently across letters. The resulting structured dataset was then imported into SAS®, where a reproducible analytical pipeline was used for data transformation, summary reporting, statistical modeling, and development of outputs for an interactive dashboard.

DATA MANAGEMENT AND WORKFLOW

DATA PREPARATION

Complete Response Letters were obtained from the openFDA Complete Response Letter dataset, which contains publicly released letters from 2002 to 2025. Because FDA application numbers may be associated with more than one CRL, a unique event identifier (event_id) was assigned to each letter to support record-level analysis.

Two structured datasets were created using coding definitions applied consistently across all reviewed letters. The first was a one-record-per-letter dataset containing application-level variables, including sponsor company, product name, review office (when available), regulatory pathway (when available), application type, and year. The second was a deficiency-level dataset stored in a long format, containing individual deficiency domains and subtypes linked to the corresponding event_id.

A deficiency-level structure was used because a single CRL may contain multiple cited deficiencies across one or more review domains. Each distinct deficiency was recorded separately under the associated event identifier. Subtypes were assigned to a single parent domain, and classification labels followed terminology used within FDA letters whenever available. This two-dataset structure supported both application-level modeling and deficiency-level frequency analyses.

Some CRLs contained redacted proprietary information. Approximately one-third of letters had missing or redacted values for variables such as regulatory pathway, office, or deficiency subtype detail. When domain headers were available, but a specific detail was removed, the domain was retained, and the subtype was coded as “Unspecified”. Redacted application-level variables were recorded accordingly in the letter-level dataset.

In total, 401 CRLs were coded, representing 916 individual deficiencies across all letters.

SAS WORKFLOW

CRL data were initially logged in Microsoft Excel and imported into SAS for processing. Permanent SAS datasets were created from the source workbook to preserve the original Excel file for future updates and provide a stable analytical environment. This process produced two primary tables: CRL_Header, containing one record per letter with application-level variables, and CRL_Deficiency, containing domain and subtype deficiency records.

The two tables were merged by event_id to create CRL_Long, the primary analytical dataset used for summaries, reporting, and downstream analyses. This merged structure allowed each CRL to retain application-level information while supporting multiple associated deficiencies.

Additional purpose-specific datasets were derived from CRL_Long as needed. For example, Deficiency_Model was created for co-occurrence analyses and logistic regression modeling, while Sample_Long was used for reporting examples and paper outputs.

ANALYTICAL METHODS

Multiple SAS procedures were used to prepare, summarize, and analyze deficiency patterns across CRLs. After the source data were imported and cleaned, PROC SORT and the DATA step processing were used to organize records by event_id, merge application-level and deficiency-level data, and create analysis-ready datasets.

PROC FREQ was used extensively to summarize categorical variables and evaluate the distribution of deficiency domains and subtypes overall, by application type, and within selected review areas. Cross-tabulations and chi-square tests were used to compare deficiency patterns across groups, such as NDA versus BLA submissions. An example of the SAS® code used for these analyses is shown below:

```
proc freq data = sasdata.crl_long order = freq;
    where deficiency_domain = 'Nonclinical';
    tables deficiency_domain*deficiency_subtype;
run;
```

Because individual CRLs may contain multiple cited deficiencies, CRL-level indicator variables were created through DATA step programming to capture whether each major review domain was present within a letter. These indicators were then used to assess co-occurrence patterns at both the domain and CMC subtype levels. Frequency outputs were further filtered to retain the more common co-occurrence patterns for reporting.

PROC LOGISTIC was used to model selected binary outcomes. Separate models were developed to evaluate whether a CRL contained a CMC deficiency and whether a CRL involved multiple deficiency domains. Predictor variables included application type, submission year, and regulatory pathway.

Additional procedures supported reporting and output preparation. PROC SQL and PROC SURVEYSELECT were used to create sample datasets for project documentation and GitHub materials,

while PROC EXPORT was used to create CSV files for downstream use in the R Shiny dashboard. Final tables and analytical reports were formatted through ODS PDF output.

RESULTS

DOMAIN FREQUENCIES

PROC FREQ was used to evaluate the distribution of deficiency domains cited in Complete Response Letters (CRLs). Chemistry, Manufacturing, and Controls was the most frequently observed domain, appearing 314 times and representing 51.5% of all coded domain findings. Clinical was the second most common domain, with 147 occurrences (24.1%). Regulatory and Nonclinical deficiencies each represented approximately 10% of findings, while Statistics and Patent Certifications were comparatively uncommon. The overall distribution of coded deficiency domains is shown in Figure 1.

Within major domains, the most common subtype findings included Facility within CMC, Efficacy within Clinical, Insufficient Nonclinical Support within Nonclinical, and Labeling within Regulatory.

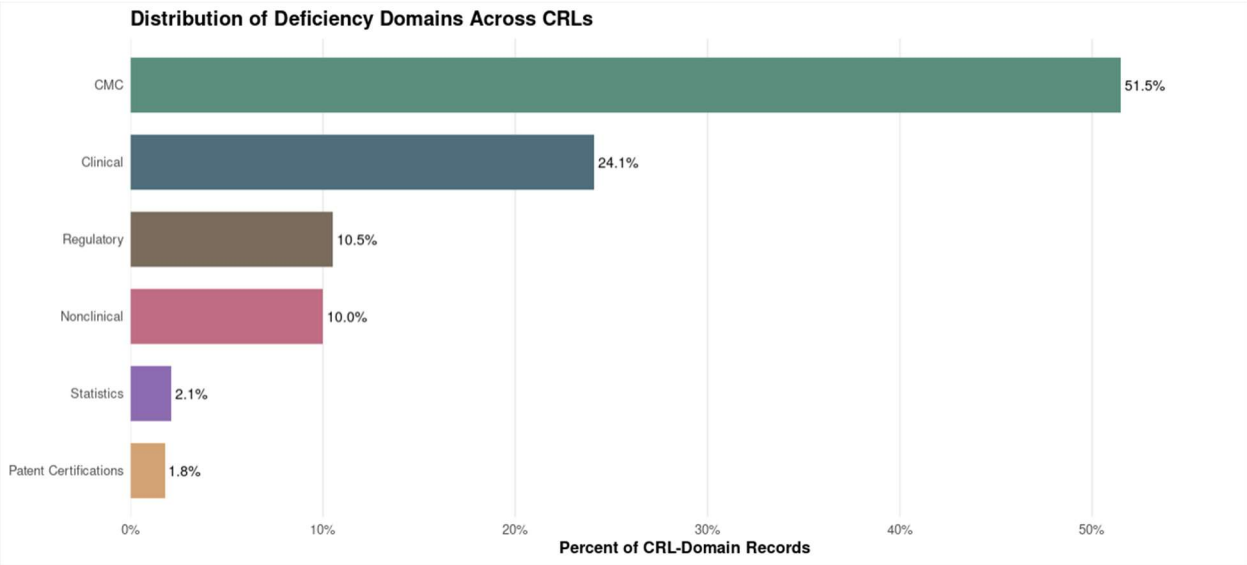

Figure 1. Distribution of Deficiency Domains Across Complete Response Letters

NDA VS BLA DIFFERENCES

Deficiency patterns were compared by application type (NDA versus BLA), as shown in table 1. Biologic License Applications showed a substantially higher proportion of CMC findings (69.2%) compared with New Drug Applications, where CMC represented 47.3% of observed domain findings. NDAs showed relatively higher proportions of Clinical and Nonclinical findings.

Overall domain distributions differed significantly between application types (Pearson chi-square = 25.92, df = 5, $p < 0.0001$).

Table 1. Distribution of Deficiency Domains by Application Type (NDA vs BLA)

Table of application_type by deficiency_domain							
	deficiency_domain(deficiency_domain)						
application_type(application_type)	CMC	Clinical	Regulatory	Nonclinical	Statistics	Patent Certifications	Total
NDA	233 47.26	125 25.35	54 10.95	60 12.17	10 2.03	11 2.23	493
BLA	81 69.23	22 18.80	10 8.55	1 0.85	3 2.56	0 0.00	117
Total	314	147	64	61	13	11	610

CO-OCCURRENCE FINDINGS

Co-occurrence patterns were evaluated to better understand how deficiency domains clustered within individual Complete Response Letters, as shown in Table 2. The most common single-domain pattern was CMC only (42.9%), followed by Clinical only (12.2%). The most common multi-domain combination was CMC + Clinical (12.7%), followed by CMC + Regulatory (7.2%). Overall, nearly 40% of CRLs contained deficiencies spanning more than one domain, highlighting the frequency of multidomain review concerns.

Table 2. Most Common Deficiency Domain Co-Occurrence Patterns

pattern	COUNT	PERCENT
CMC	172	42.8928
CMC, Clinical	51	12.7182
Clinical	49	12.2195
CMC, Regulatory	27	6.7332
CMC, Nonclinical	21	5.2369
CMC, Clinical, Nonclinical	11	2.7431

LOGISTIC REGRESSION MODELS

Logistic Model 1 (CMC)

PROC LOGISTIC was used to evaluate predictors of whether a CRL contained a CMC deficiency. In the multivariable model, the regulatory pathway was significantly associated with CMC findings (Type 3 $p = 0.0069$), while application type and year were not statistically significant after adjustment.

Relative to 505(b)(2) applications, 505(b) submissions had significantly lower odds of containing a CMC deficiency (OR = 0.343, 95% CI: 0.171–0.687). Model discrimination was moderate ($c = 0.645$).

Logistic Model 2 (Multi-domain)

A second logistic regression model evaluated predictors of whether a CRL involved more than one deficiency domain. Application type, year, and regulatory pathway were all significant predictors ($p < 0.01$).

BLAs had substantially lower odds of multi-domain CRLs compared with NDAs (OR = 0.051, 95% CI: 0.011–0.227). Later submission years were associated with lower odds of multi-domain findings (OR = 0.893 per year, 95% CI: 0.828–0.963). Model discrimination was acceptable (c = 0.668).

DASHBOARD REPORTING

An interactive dashboard was developed in R Shiny to complement the SAS reporting workflow and allow dynamic exploration of Complete Response Letter trends. Users can filter and visualize deficiency domains, subtype frequencies, temporal trends, and common co-occurrence patterns. This platform provides an accessible method for reviewing results beyond static tables and figures. A screenshot of the dashboard interface is shown in Figure 2.

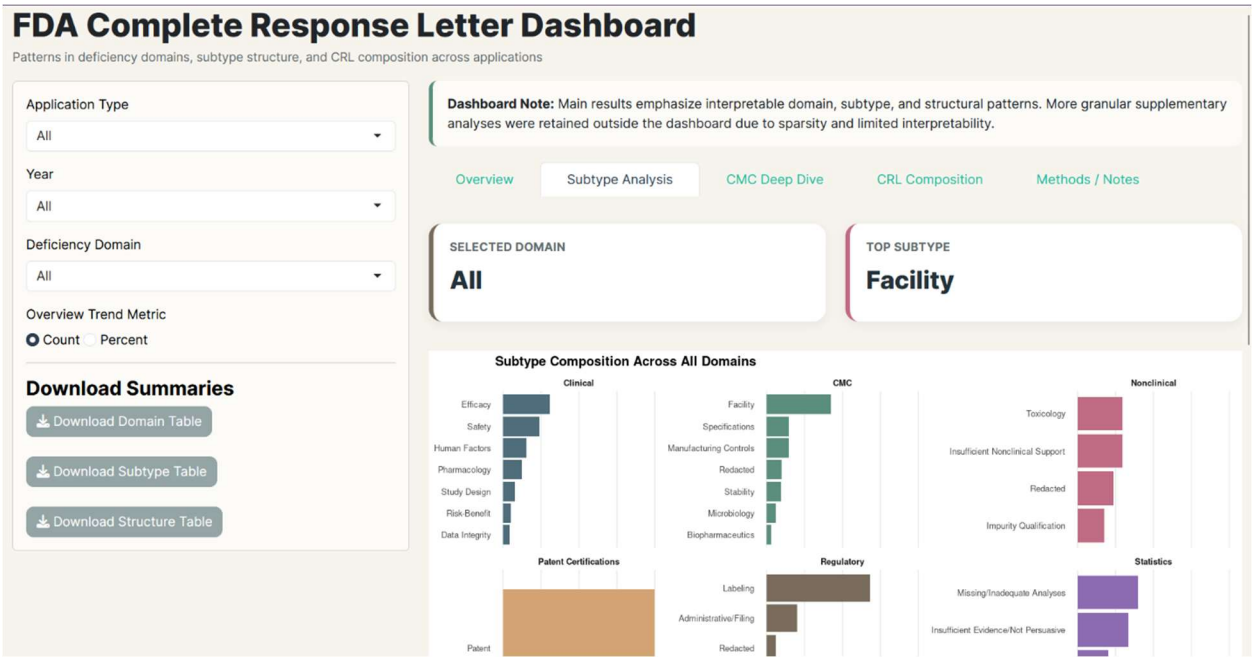

Figure 2. Interactive Dashboard for Exploring Complete Response Letter Deficiency Patterns. Live version available at: https://hannahweber-crl-deficiency-analysis.share.connect.posit.cloud/

DISCUSSION

The predominance of CMC deficiencies suggests that manufacturing readiness, process controls, and facility compliance remain major barriers to regulatory approval. This finding was especially pronounced among biologic applications, where production complexity and quality assurance requirements are often greater.

Differences observed across regulatory pathways may reflect varying evidentiary expectations, product novelty, or reliance on prior approvals. These patterns indicate that application type and submission strategy may influence the types of deficiencies identified during review.

The frequency of multi-domain CRLs demonstrates that many non-approvals involve multiple concurrent issues rather than a single isolated deficiency. This highlights the importance of integrated development strategies spanning clinical, manufacturing, and regulatory functions.

Beyond the findings themselves, this project demonstrates how SAS can be used to transform unstructured regulatory documents into analyzable datasets suitable for repeated reporting and further research.

LIMITATIONS

Several limitations should be considered when interpreting these findings. First, the analysis was limited to publicly released Complete Response Letters selected for disclosure by the FDA, which may not represent the full population of issued CRLs and could introduce selection bias. Second, this was an observational study based on publicly available regulatory documents, and portions of some letters were intentionally redacted. Missing information related to variables such as regulatory pathway, office, or subtype detail may have reduced precision and affected selected comparisons. Third, because CRLs were manually reviewed and coded into structured variables, some misclassification or human coding error is possible despite efforts to apply definitions consistently across letters.

CONCLUSION

This project demonstrates that SAS can successfully transform publicly available Complete Response Letters into structured datasets suitable for statistical analysis, reporting, and visualization. CMC deficiencies were the most common findings, meaningful differences were observed across application types, and many CRLs involved multiple review domains.

The workflow provides a practical framework for future regulatory analytics using publicly available FDA data. Future enhancements could incorporate artificial intelligence or natural language processing methods to automate CRL coding, reduce manual review burden, and expand large-scale analysis of regulatory deficiency trends.

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